

Antiderivatives

the problem is to find a function F whose derivative is a known function f .

If such a function F exists, it is called an *anti-derivative* of f .

Definition A function F is called an **antiderivative** of f on an interval I if $F'(x) = f(x)$ for all x in I .

For instance, let $f(x) = x^2$. It isn't difficult to discover an antiderivative of f if we keep the Power Rule in mind. In fact, if $F(x) = \frac{1}{3}x^3$, then $F'(x) = x^2 = f(x)$. But the function $G(x) = \frac{1}{3}x^3 + 100$ also satisfies $G'(x) = x^2$. Therefore, both F and G are antiderivatives of f . Indeed, any function of the form $H(x) = \frac{1}{3}x^3 + C$, where C is a constant, is an antiderivative of f . The question arises: Are there any others?

if F and G are any two antiderivatives of f , then $F'(x) = f(x) = G'(x)$

so $G(x) - F(x) = C$, where C is a constant. We can write this as $G(x) = F(x) + C$,

Theorem If F is an antiderivative of f on an interval I , then the most general antiderivative of f on I is

$$F(x) + C$$

where C is an arbitrary constant.

EXAMPLE 1 Find the most general antiderivative of each of the following functions.

(a) $f(x) = \sin x$ (b) $f(x) = 1/x$ (c) $f(x) = x^n, \quad n \neq -1$

SOLUTION

(a) If $F(x) = -\cos x$, then $F'(x) = \sin x$, so an antiderivative of $\sin x$ is $-\cos x$. By Theorem , the most general antiderivative is $G(x) = -\cos x + C$.

(b) Recall

$$\frac{d}{dx} (\ln x) = \frac{1}{x}$$

So on the interval $(0, \infty)$ the general antiderivative of $1/x$ is $\ln x + C$. We also learned that

$$\frac{d}{dx} (\ln |x|) = \frac{1}{x}$$

for all $x \neq 0$. Theorem then tells us that the general antiderivative of $f(x) = 1/x$ is $\ln |x| + C$ on any interval that doesn't contain 0. In particular, this is true on each of the intervals $(-\infty, 0)$ and $(0, \infty)$. So the general antiderivative of f is

$$F(x) = \begin{cases} \ln x + C_1 & \text{if } x > 0 \\ \ln(-x) + C_2 & \text{if } x < 0 \end{cases}$$

(c) We use the Power Rule to discover an antiderivative of x^n . In fact, if $n \neq -1$, then

$$\frac{d}{dx} \left(\frac{x^{n+1}}{n+1} \right) = \frac{(n+1)x^n}{n+1} = x^n$$

Thus, the general antiderivative of $f(x) = x^n$ is

$$F(x) = \frac{x^{n+1}}{n+1} + C$$

This is valid for $n \geq 0$ since then $f(x) = x^n$ is defined on an interval. If n is negative (but $n \neq -1$), it is valid on any interval that doesn't contain 0. 

Function	Particular antiderivative	Function	Particular antiderivative
$cf(x)$	$cF(x)$	$\sin x$	$-\cos x$
$f(x) + g(x)$	$F(x) + G(x)$	$\sec^2 x$	$\tan x$
$x^n (n \neq -1)$	$\frac{x^{n+1}}{n+1}$	$\sec x \tan x$	$\sec x$
$1/x$	$\ln x $	$\frac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x$
e^x	e^x	$\frac{1}{1+x^2}$	$\tan^{-1} x$
$\cos x$	$\sin x$		

EXAMPLE 2 Find all functions g such that

$$g'(x) = 4 \sin x + \frac{2x^5 - \sqrt{x}}{x}$$

SOLUTION We first rewrite the given function as follows:

$$g'(x) = 4 \sin x + \frac{2x^5}{x} - \frac{\sqrt{x}}{x} = 4 \sin x + 2x^4 - \frac{1}{\sqrt{x}}$$

Thus, we want to find an antiderivative of

$$g'(x) = 4 \sin x + 2x^4 - x^{-1/2}$$

Using the formulas in Table

$$\begin{aligned} g(x) &= 4(-\cos x) + 2 \frac{x^5}{5} - \frac{x^{1/2}}{\frac{1}{2}} + C \\ &= -4 \cos x + \frac{2}{5} x^5 - 2\sqrt{x} + C \end{aligned}$$

EXAMPLE 3 Find f if $f'(x) = e^x + 20(1 + x^2)^{-1}$ and $f(0) = -2$.

SOLUTION The general antiderivative of

$$f'(x) = e^x + \frac{20}{1 + x^2}$$

is $f(x) = e^x + 20 \tan^{-1}x + C$

To determine C we use the fact that $f(0) = -2$:

$$f(0) = e^0 + 20 \tan^{-1} 0 + C = -2$$

Thus, we have $C = -2 - 0 = -2$, so the particular solution is

$$f(x) = e^x + 20 \tan^{-1}x - 2$$

EXAMPLE 4 Find f if $f''(x) = 12x^2 + 6x - 4$, $f(0) = 4$, and $f(1) = 1$.

SOLUTION The general antiderivative of $f''(x) = 12x^2 + 6x - 4$ is

$$f'(x) = 12 \frac{x^3}{3} + 6 \frac{x^2}{2} - 4x + C = 4x^3 + 3x^2 - 4x + C$$

Using the antidifferentiation rules once more, we find that

$$f(x) = 4 \frac{x^4}{4} + 3 \frac{x^3}{3} - 4 \frac{x^2}{2} + Cx + D = x^4 + x^3 - 2x^2 + Cx + D$$

To determine C and D we use the given conditions that $f(0) = 4$ and $f(1) = 1$. Since $f(0) = 0 + D = 4$, we have $D = 4$. Since

$$f(1) = 1 + 1 - 2 + C + 4 = 1$$

we have $C = -3$. Therefore, the required function is

$$f(x) = x^4 + x^3 - 2x^2 - 3x + 4$$

Indefinite Integrals

We need a convenient notation for antiderivatives that makes them easy to work with.

the notation $\int f(x) dx$ is traditionally used for an antiderivative of f and is called an indefinite integral. Thus

$$\int f(x) dx = F(x) \quad \text{means} \quad F'(x) = f(x)$$

For example, we can write

$$\int x^2 dx = \frac{x^3}{3} + C \quad \text{because} \quad \frac{d}{dx} \left(\frac{x^3}{3} + C \right) = x^2$$

1 Table of Indefinite Integrals

$$\int cf(x) dx = c \int f(x) dx$$

$$\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$$

$$\int k dx = kx + C$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$\int \frac{1}{x^2 + 1} dx = \tan^{-1}x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1}x + C$$

Some elementary integrals

1. $\int 1 \, dx = x + C$
2. $\int x \, dx = \frac{1}{2}x^2 + C$
3. $\int x^2 \, dx = \frac{1}{3}x^3 + C$
4. $\int \frac{1}{x^2} \, dx = -\frac{1}{x} + C$
5. $\int \sqrt{x} \, dx = \frac{2}{3}x^{3/2} + C$
6. $\int \frac{1}{\sqrt{x}} \, dx = 2\sqrt{x} + C$
7. $\int x^r \, dx = \frac{1}{r+1}x^{r+1} + C \quad (r \neq -1)$
8. $\int \frac{1}{x} \, dx = \ln|x| + C$
9. $\int \sin ax \, dx = -\frac{1}{a} \cos ax + C$
10. $\int \cos ax \, dx = \frac{1}{a} \sin ax + C$
11. $\int \sec^2 ax \, dx = \frac{1}{a} \tan ax + C$
12. $\int \csc^2 ax \, dx = -\frac{1}{a} \cot ax + C$
13. $\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax + C$
14. $\int \csc ax \cot ax \, dx = -\frac{1}{a} \csc ax + C$
15. $\int \frac{1}{\sqrt{a^2 - x^2}} \, dx = \sin^{-1} \frac{x}{a} + C \quad (a > 0)$
16. $\int \frac{1}{a^2 + x^2} \, dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$
17. $\int e^{ax} \, dx = \frac{1}{a} e^{ax} + C$
18. $\int b^{ax} \, dx = \frac{1}{a \ln b} b^{ax} + C$
19. $\int \cosh ax \, dx = \frac{1}{a} \sinh ax + C$
20. $\int \sinh ax \, dx = \frac{1}{a} \cosh ax + C$

$$\int \frac{dx}{ax+b} = \frac{1}{a} \ln |ax+b| + C$$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \ln \frac{x-a}{x+a} + C$$

$$\int \tan ax \, dx = \frac{1}{a} \ln \sec ax + C$$

$$\int \cot ax \, dx = \frac{1}{a} \ln \sin ax + C$$

$$\int \sec x \, dx = \ln (\sec x + \tan x) + C$$

$$\int \operatorname{cosec} x \, dx = \ln (\operatorname{cosec} x - \cot x) + C$$

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln (x + \sqrt{x^2+a^2}) + C$$

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \ln (x + \sqrt{x^2-a^2}) + C$$

The linearity formula

$$\int (A f(x) + B g(x)) dx = A \int f(x) dx + B \int g(x) dx$$

makes it possible to integrate sums and constant multiples of functions.

EXAMPLE (Combining elementary integrals)

$$(a) \int (x^4 - 3x^3 + 8x^2 - 6x - 7) dx = \frac{x^5}{5} - \frac{3x^4}{4} + \frac{8x^3}{3} - 3x^2 - 7x + C$$

$$(b) \int \left(5x^{3/5} - \frac{3}{2+x^2} \right) dx = \frac{25}{8}x^{8/5} - \frac{3}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} + C$$

$$(c) \int (4 \cos 5x - 5 \sin 3x) dx = \frac{4}{5} \sin 5x + \frac{5}{3} \cos 3x + C$$

$$(d) \int \left(\frac{1}{\pi x} + a^{\pi x} \right) dx = \frac{1}{\pi} \ln |x| + \frac{1}{\pi \ln a} a^{\pi x} + C, \quad (a > 0).$$

EXAMPLE

$$\begin{aligned}\int \frac{(x+1)^3}{x} dx &= \int \frac{x^3 + 3x^2 + 3x + 1}{x} dx \\ &= \int \left(x^2 + 3x + 3 + \frac{1}{x} \right) dx \\ &= \frac{1}{3}x^3 + \frac{3}{2}x^2 + 3x + \ln|x| + C.\end{aligned}$$

We adopt the convention that

when a formula for a general indefinite integral is given, it is valid only on an interval. Thus, we write

$$\int \frac{1}{x^2} dx = -\frac{1}{x} + C$$

with the understanding that it is valid on the interval $(0, \infty)$ or on the interval $(-\infty, 0)$. This is true despite the fact that the general antiderivative of the function $f(x) = 1/x^2$, $x \neq 0$, is

$$F(x) = \begin{cases} -\frac{1}{x} + C_1 & \text{if } x < 0 \\ -\frac{1}{x} + C_2 & \text{if } x > 0 \end{cases}$$

EXAMPLE Find the general indefinite integral

$$\int (10x^4 - 2 \sec^2 x) dx$$

SOLUTION Using our convention and Table we have

$$\begin{aligned}\int (10x^4 - 2 \sec^2 x) dx &= 10 \int x^4 dx - 2 \int \sec^2 x dx \\ &= 10 \frac{x^5}{5} - 2 \tan x + C \\ &= 2x^5 - 2 \tan x + C\end{aligned}$$

EXAMPLE Evaluate $\int \frac{\cos \theta}{\sin^2 \theta} d\theta$.

SOLUTION This indefinite integral isn't immediately apparent in Table 1, so we use trigonometric identities to rewrite the function before integrating:

$$\begin{aligned}\int \frac{\cos \theta}{\sin^2 \theta} d\theta &= \int \left(\frac{1}{\sin \theta} \right) \left(\frac{\cos \theta}{\sin \theta} \right) d\theta \\ &= \int \csc \theta \cot \theta d\theta = -\csc \theta + C\end{aligned}$$

The most important of these techniques is

the **method of substitution**, the integral version of the Chain Rule. If we rewrite the Chain Rule, $\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$, in integral form, we obtain

$$\int f'(g(x)) g'(x) dx = f(g(x)) + C.$$

Observe that the following formalism would produce this latter formula even if we did not already know it was true:

Let $u = g(x)$. Then $du/dx = g'(x)$, or in differential form, $du = g'(x) dx$. Thus,

$$\int f'(g(x)) g'(x) dx = \int f'(u) du = f(u) + C = f(g(x)) + C.$$

if $F' = f$, then

$$\int F'(g(x))g'(x) dx = F(g(x)) + C$$

because, by the Chain Rule,

$$\frac{d}{dx} [F(g(x))] = F'(g(x))g'(x)$$

If we make the “change of variable” or “substitution” $u = g(x)$,

we have

$$\int F'(g(x))g'(x) dx = F(g(x)) + C = F(u) + C = \int F'(u) du$$

or, writing $F' = f$, we get

$$\int f(g(x))g'(x) dx = \int f(u) du$$

The Substitution Rule If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I , then

$$\int f(g(x))g'(x) dx = \int f(u) du$$

EXAMPLE Find $\int x^3 \cos(x^4 + 2) dx$.

SOLUTION We make the substitution $u = x^4 + 2$ because its differential is $du = 4x^3 dx$, which, apart from the constant factor 4, occurs in the integral. Thus, using $x^3 dx = du/4$ and the Substitution Rule, we have

$$\begin{aligned}\int x^3 \cos(x^4 + 2) dx &= \int \cos u \cdot \frac{1}{4} du = \frac{1}{4} \int \cos u du \\ &= \frac{1}{4} \sin u + C \\ &= \frac{1}{4} \sin(x^4 + 2) + C\end{aligned}$$

Notice that at the final stage we had to return to the original variable x .

EXAMPLE Evaluate $\int \sqrt{2x+1} \, dx$.

SOLUTION 1 Let $u = 2x + 1$. Then $du = 2 \, dx$, so $dx = du/2$. Thus, the Substitution Rule gives

$$\begin{aligned}\int \sqrt{2x+1} \, dx &= \int \sqrt{u} \frac{du}{2} = \frac{1}{2} \int u^{1/2} \, du \\ &= \frac{1}{2} \cdot \frac{u^{3/2}}{3/2} + C = \frac{1}{3} u^{3/2} + C \\ &= \frac{1}{3} (2x+1)^{3/2} + C\end{aligned}$$

SOLUTION 2 Another possible substitution is $u = \sqrt{2x+1}$. Then

$$du = \frac{dx}{\sqrt{2x+1}} \quad \text{so} \quad dx = \sqrt{2x+1} \, du = u \, du$$

(Or observe that $u^2 = 2x + 1$, so $2u \, du = 2 \, dx$.) Therefore

$$\begin{aligned}\int \sqrt{2x+1} \, dx &= \int u \cdot u \, du = \int u^2 \, du \\ &= \frac{u^3}{3} + C = \frac{1}{3} (2x+1)^{3/2} + C\end{aligned}$$

EXAMPLE Find $\int \frac{x}{\sqrt{1-4x^2}} dx$.

SOLUTION Let $u = 1 - 4x^2$. Then $du = -8x dx$, so $x dx = -\frac{1}{8} du$ and

$$\begin{aligned}\int \frac{x}{\sqrt{1-4x^2}} dx &= -\frac{1}{8} \int \frac{du}{\sqrt{u}} = -\frac{1}{8} \int u^{-1/2} du \\ &= -\frac{1}{8}(2\sqrt{u}) + C = -\frac{1}{4}\sqrt{1-4x^2} + C\end{aligned}$$

EXAMPLE Calculate $\int e^{5x} dx$.

SOLUTION If we let $u = 5x$, then $du = 5 dx$, so $dx = \frac{1}{5} du$. Therefore

$$\int e^{5x} dx = \frac{1}{5} \int e^u du = \frac{1}{5} e^u + C = \frac{1}{5} e^{5x} + C$$

EXAMPLE Find $\int \sqrt{1 + x^2} x^5 dx$.

SOLUTION An appropriate substitution becomes more obvious if we factor x^5 as $x^4 \cdot x$. Let $u = 1 + x^2$. Then $du = 2x dx$, so $x dx = du/2$. Also $x^2 = u - 1$, so $x^4 = (u - 1)^2$:

$$\begin{aligned}\int \sqrt{1 + x^2} x^5 dx &= \int \sqrt{1 + x^2} x^4 \cdot x dx \\&= \int \sqrt{u} (u - 1)^2 \frac{du}{2} = \frac{1}{2} \int \sqrt{u} (u^2 - 2u + 1) du \\&= \frac{1}{2} \int (u^{5/2} - 2u^{3/2} + u^{1/2}) du \\&= \frac{1}{2} \left(\frac{2}{7} u^{7/2} - 2 \cdot \frac{2}{5} u^{5/2} + \frac{2}{3} u^{3/2} \right) + C \\&= \frac{1}{7} (1 + x^2)^{7/2} - \frac{2}{5} (1 + x^2)^{5/2} + \frac{1}{3} (1 + x^2)^{3/2} + C\end{aligned}$$

EXAMPLE Calculate $\int \tan x \, dx$.

SOLUTION First we write tangent in terms of sine and cosine:

$$\int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx$$

This suggests that we should substitute $u = \cos x$, since then $du = -\sin x \, dx$ and so $\sin x \, dx = -du$:

$$\begin{aligned} \int \tan x \, dx &= \int \frac{\sin x}{\cos x} \, dx = -\int \frac{du}{u} \\ &= -\ln |u| + C = -\ln |\cos x| + C \end{aligned}$$

Since $-\ln |\cos x| = \ln(|\cos x|^{-1}) = \ln(1/|\cos x|) = \ln |\sec x|$, the result of Example 1 can also be written as

$$\int \tan x \, dx = \ln |\sec x| + C$$

EXAMPLE

(Examples of substitution) Find the indefinite integrals:

(a) $\int \frac{x}{x^2 + 1} dx$, (b) $\int \frac{\sin(3 \ln x)}{x} dx$, and (c) $\int e^x \sqrt{1 + e^x} dx$.

Solution

(a) $\int \frac{x}{x^2 + 1} dx$ Let $u = x^2 + 1$.
Then $du = 2x dx$ and
 $x dx = \frac{1}{2} du$
$$= \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln |u| + C = \frac{1}{2} \ln(x^2 + 1) + C = \ln \sqrt{x^2 + 1} + C.$$

(Both versions of the final answer are equally acceptable.)

(b) $\int \frac{\sin(3 \ln x)}{x} dx$ Let $u = 3 \ln x$.
Then $du = \frac{3}{x} dx$
$$= \frac{1}{3} \int \sin u du = -\frac{1}{3} \cos u + C = -\frac{1}{3} \cos(3 \ln x) + C.$$

(c) $\int e^x \sqrt{1 + e^x} dx$ Let $v = 1 + e^x$.
Then $dv = e^x dx$
$$= \int v^{1/2} dv = \frac{2}{3} v^{3/2} + C = \frac{2}{3} (1 + e^x)^{3/2} + C.$$

it may be necessary to manipulate the integrand algebraically to put it into a better form for substitution.

EXAMPLE

Evaluate (a) $\int \frac{1}{x^2 + 4x + 5} dx$ and (b) $\int \frac{dx}{\sqrt{e^{2x} - 1}}$.

Solution

$$\begin{aligned} \text{(a)} \quad \int \frac{dx}{x^2 + 4x + 5} &= \int \frac{dx}{(x + 2)^2 + 1} && \text{Let } t = x + 2. \\ &&& \text{Then } dt = dx. \\ &= \int \frac{dt}{t^2 + 1} \\ &= \tan^{-1} t + C = \tan^{-1}(x + 2) + C. \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \int \frac{dx}{\sqrt{e^{2x} - 1}} &= \int \frac{dx}{e^x \sqrt{1 - e^{-2x}}} \\ &= \int \frac{e^{-x} dx}{\sqrt{1 - (e^{-x})^2}} && \text{Let } u = e^{-x}. \\ &&& \text{Then } du = -e^{-x} dx. \\ &= - \int \frac{du}{\sqrt{1 - u^2}} \\ &= -\sin^{-1} u + C = -\sin^{-1}(e^{-x}) + C. \end{aligned}$$

$$\textcircled{1} \int (x^2+x-1)^4 \cdot (2x+1) dx = \int u^4 \cdot du = \frac{u^5}{5} + C$$

$$x^2+x-1=u$$

$$(2x+1) dx = du$$

$$= \frac{(x^2+x-1)^5}{5} + C$$

$$\textcircled{2} \int \sqrt{2x+1} dx = \int \sqrt{u} \cdot \frac{du}{2} = \frac{1}{2} \cdot \int \sqrt{u} du = \frac{1}{2} \int u^{1/2} du$$

$$2x+1=u$$

$$2 dx = du \Rightarrow dx = \frac{du}{2}$$

$$= \frac{1}{2} \frac{u^{3/2}}{3/2} = \frac{1}{3} \sqrt{(2x+1)^3} + C$$

$$\textcircled{3} \int \cos(3x+1) dx = \int \cos(u) \cdot \frac{du}{3} = \frac{1}{3} \int \cos u du$$

$$3x+1=u$$

$$3dx=du$$

$$= \frac{1}{3} \cdot \sin u + C = \frac{1}{3} \sin(3x+1) + C$$

$$\textcircled{4} \int x^2 \cdot \sin(x^3) dx = \int \sin u \cdot \frac{1}{3} du = -\frac{1}{3} \cos u + C$$

$$x^3=u \quad 3x^2 dx = du = -\frac{1}{3} \cos x^3 + C$$

$$\textcircled{5} \int x \sqrt{x^2+1} dx = \int \sqrt{u} \cdot \frac{1}{2} du$$

$$x^2+1=u \Rightarrow 2x dx = du$$

$$\textcircled{6} \int \frac{x dx}{\sqrt{x^2+1}} = \int \frac{du/2}{\sqrt{u}} = \frac{1}{2} \int u^{-1/2} = \frac{1}{2} \frac{u^{1/2}}{1/2} + C = \sqrt{x^2+1} + C$$

$$x^2+1=u \quad 2x dx = du$$

$$\textcircled{7} \int \frac{\ln x}{x} dx = \int u du = \frac{u^2}{2} + C = \frac{(\ln x)^2}{2} + C$$

$$\ln x = u \quad \frac{1}{x} dx = du$$

$$\textcircled{8} \int x^2 e^{x^3} dx = \int e^u \cdot \frac{du}{3} = \frac{1}{3} \cdot e^u + C$$

$$= \frac{1}{3} e^{x^3} + C$$

$$x^3 = u$$

$$3x^2 dx = du$$

$$\textcircled{9} \int \frac{dx}{\sqrt{5x-1}} = \frac{1}{5} \int \frac{du}{\sqrt{u}}$$

$$5x-1=u$$

$$5dx = du \quad dx = \frac{du}{5}$$

$$\textcircled{10} \int \tan^2 x \sec^2 x \, dx = \int u^2 \, du = \frac{u^3}{3} + C = \frac{\tan^3 x}{3} + C$$

$\tan x = u$
 $\sec^2 x \, dx = du$
 $\left(\frac{\sin x}{\cos x}\right)' = \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$

$$\textcircled{11} \int \frac{\sec x \tan x}{\sqrt{\sec x}} \, dx = \int \frac{du}{\sqrt{u}} = \dots$$

$$\sec x = u \Rightarrow \sec x \tan x \, dx = du$$

$$\left(\frac{1}{\cos x}\right)' = \frac{\sin x}{\cos^2 x}$$

$$(12) \int \frac{1}{x^2} \cdot \cos\left(\frac{1}{x}\right) dx = -\int \cos u du = -\sin u + C$$

$$= -\sin\left(\frac{1}{x}\right) + C$$

$$\frac{1}{x} = u \Rightarrow -\frac{1}{x^2} dx = du$$

$$(13) \int x(x-1)^9 dx = \int (u+1) \cdot u^9 du = \int (u^{10} + u^9) du$$

$$x-1 = u$$

$$dx = du$$

$$= \frac{u^{11}}{11} + \frac{u^{10}}{10} + C$$

$$\int e^{\cos x} \cdot \sin x dx = ? \quad \begin{array}{l} t = \cos x \\ dt = -\sin x dx \end{array}$$

$$\int e^{\cos x} \cdot \sin x dx = -\int e^t + c = -e^{\cos x} + c$$

$$\int \sqrt{\sin x} \cdot \cos x dx = ? \quad \begin{array}{l} t = \sin x \\ dt = \cos x dx \end{array}$$

$$\int \sqrt{\sin x} \cdot \cos x dx = \int \sqrt{t} \cdot dt = \frac{2}{3} t^{3/2} + c = \frac{2}{3} \sin^{3/2} x + c$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = ? \quad \begin{array}{l} t = \frac{x}{a} \Rightarrow x = t \cdot a \\ dx = a dt \end{array}$$

$$\begin{aligned} \int \frac{dx}{\sqrt{a^2 - x^2}} &= \frac{1}{a} \int \frac{dx}{\sqrt{1 - (\frac{x}{a})^2}} = \frac{1}{a} \int \frac{a dt}{\sqrt{1 - t^2}} = \arcsin t + c \\ &= \arcsin \frac{x}{a} + c \end{aligned}$$



POWER RULE

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c \quad (n \neq -1)$$

$$\int u^n dx = \frac{u^{n+1}}{n+1} + c \quad (n \neq -1) \text{ for any } u$$

Ex. Find $\int (x+1)^{20} dx$

set $u = x+1$
 $du = dx$

$$\begin{aligned} \int (x+1)^{20} dx &= \int u^{20} du = \frac{u^{21}}{21} + c \\ &= \frac{(x+1)^{21}}{21} + c \end{aligned}$$

* Ex: Find $\int 3x^2(x^3+7)^3 dx$

Let $u = x^3 + 7$
 $du = 3x^2 dx$

$$\begin{aligned}\int \underbrace{3x^2}_{du} \underbrace{(x^3+7)^3}_{u^3} dx &= \int u^3 \cdot du \\ &= \frac{u^4}{4} + C \\ &= \frac{(x^3+7)^4}{4} + C\end{aligned}$$

* Ex: $\int x \cdot \sqrt{x^2+5} dx$

Let $u = x^2 + 5$
 $du = 2x dx$
 $\frac{du}{2} = x dx$

$$\begin{aligned}\int x \sqrt{x^2+5} dx &= \int \underbrace{x}_{\frac{du}{2}} \cdot \underbrace{(x^2+5)^{1/2}}_{u^{1/2}} dx \\ &= \int u^{1/2} \cdot \frac{du}{2} = \frac{1}{2} \int u^{1/2} du \\ &= \frac{1}{2} \cdot \frac{u^{3/2}}{\frac{3}{2}} + C = \frac{1}{3} u^{3/2} + C \\ &= \frac{1}{3} (x^2+5)^{3/2} + C\end{aligned}$$

$$\begin{aligned}
 \text{Ex } \int \sqrt[3]{6y} \, dy &= \int 6^{1/3} \cdot y^{1/3} \, dy \\
 &= \sqrt[3]{6} \int y^{1/3} \, dy = \sqrt[3]{6} \frac{y^{4/3}}{\frac{4}{3}} + C \\
 &= \frac{\sqrt[3]{6}}{4} y^{4/3} + C
 \end{aligned}$$

$$\begin{aligned}
 * \text{Ex: } \int \frac{2x^3 + 3x}{(x^4 + 3x^2 + 7)^4} \, dx &= \int \frac{du/2}{u^4} = \int \frac{u^{-4} \, du}{2} \\
 &= \frac{1}{2} \int u^{-4} \, du = \frac{1}{2} \frac{u^{-3}}{-3} + C
 \end{aligned}$$

$$x^4 + 3x^2 + 7 = u$$

$$(4x^3 + 6x) \, dx = du$$

$$2(2x^3 + 3x) \, dx = du$$

$$(2x^3 + 3x) \, dx = \frac{du}{2}$$

$$= -\frac{1}{6u^3} + C$$

$$= -\frac{1}{6(x^4 + 3x^2 + 7)^3} + C$$

EX: $I = \int x (x^2 + 2)^3 dx = ?$

$$x^2 + 2 = u$$

$$2x dx = du$$

$$x dx = \frac{du}{2}$$

$$\int \underbrace{x (x^2 + 2)^3 dx}_{\text{circled}} = \int u^3 \cdot \frac{du}{2}$$

$$= \frac{u^4}{\cancel{8}} + C$$

$$= \frac{(x^2 + 2)^4}{8} + C$$

EX: $I = \int \frac{x+1}{(x^2 + 2x + 5)^2} dx = ?$

$$x^2 + 2x + 5 = u$$

$$(2x + 2) dx = du$$

$$2(x + 1) dx = du$$

$$(x + 1) dx = \frac{du}{2}$$

$$\int \frac{x+1}{(x^2 + 2x + 5)^2} dx = \int \frac{\frac{du}{2}}{u^2} = \frac{1}{2} \int u^{-2} du$$

$$= \frac{1}{2} \frac{u^{-1}}{-1} + C = -\frac{1}{2u} + C$$

$$= -\frac{1}{2(x^2 + 2x + 5)} + C$$

Ex: $I = \int \frac{6x}{(2x^2-7)^{10}} dx = 6 \int \frac{du/4}{u^{10}}$

$$u = 2x^2 - 7$$

$$du = 4x dx$$

$$\frac{du}{4} = x dx$$

$$= \frac{6}{4} \int u^{-10} du$$

$$= \frac{6}{4} \frac{u^{-9}}{-9} + C$$

$$= \frac{3}{2} \frac{(2x^2-7)^{-9}}{-9} + C$$

$$I = \int (2x+4) (3x^2+12x)^{3/2} dx$$

$$u = 3x^2 + 12x$$

$$du = (6x+12) dx$$

$$du = 3(2x+4) dx$$

$$\frac{du}{3} = (2x+4) dx$$

$$\begin{aligned} & \int \underline{(2x+4)} (6x^2+12x)^{3/2} \underline{dx} = \\ &= \int u^{3/2} \cdot \frac{du}{3} = \frac{1}{3} \frac{u^{5/2}}{5/2} + C \\ &= \frac{2}{15} (3x^2+12x)^{5/2} + C \end{aligned}$$

$$\int e^u du = e^u + c$$

Ex: Find $\int 2x e^{x^2} dx = ?$

$$\begin{aligned} x^2 &= u \\ 2x dx &= du \end{aligned} \quad \int 2x e^{x^2} dx = \int e^u du = e^u + c = e^{x^2} + c$$

Ex: $\int (x^2+1) e^{x^3+3x} dx = ?$

$$\begin{aligned} x^3+3x &= u \\ (3x^2+3) dx &= du \\ 3(x^2+1) dx &= du \\ (x^2+1) dx &= \frac{du}{3} \end{aligned} \quad \int (x^2+1) e^{x^3+3x} dx = \int e^u \cdot \frac{du}{3} = \frac{1}{3} e^u + c = \frac{1}{3} e^{(x^3+3x)} + c$$

Ex: $\int (x^2 + 3 - x \cdot e^{7-x^2}) dx = ?$

$$\int (x^2 + 3) dx - \int x e^{7-x^2} dx$$

$$7 - x^2 = u$$

$$-2x dx = du$$

$$\frac{x^3}{3} + 3x - \int e^u \left(-\frac{du}{2}\right)$$

$$x dx = -\frac{du}{2}$$

$$\frac{x^3}{3} + 3x + \frac{1}{2} \int e^u du = \frac{x^3}{3} + 3x + \frac{1}{2} e^u + C$$

$$= \frac{x^3}{3} + 3x + \frac{1}{2} e^{7-x^2} + C$$

$$\int \frac{1}{u} du = \ln |u| + C \quad \text{for } u \neq 0$$

Ex: Find $\int \frac{2x}{x^2+5} dx = ?$

$$\begin{aligned} x^2+5 &= u \\ 2x dx &= du \end{aligned} \quad \int \frac{2x}{x^2+5} dx = \int \frac{du}{u} = \ln |u| + C = \ln |x^2+5| + C$$

Since x^2+5 is always positive, we can omit the absolute value bars:

$$\int \frac{2x}{x^2+5} dx = \ln(x^2+5) + C$$

$$\int \frac{(2x^3 + 3x) dx}{x^4 + 3x^2 + 7}$$

$$u = x^4 + 3x^2 + 7$$

$$du = (4x^3 + 6x) dx$$

$$du = 2(2x^3 + 3x) dx$$

$$\frac{du}{2} = (2x^3 + 3x) dx$$

$$\int \frac{(2x^3 + 3x) dx}{x^4 + 3x^2 + 7} = \int \frac{du/2}{u}$$

$$= \frac{1}{2} \ln|u| + C$$

$$= \frac{1}{2} \ln|x^4 + 3x^2 + 7| + C$$

$$= \frac{1}{2} \ln(x^4 + 3x^2 + 7) + C$$

$$x^4 + 3x^2 + 7 > 0$$

for all x

$$\text{Ex: } \int \left[\frac{1}{1-x} + \frac{1}{(1-x)^2} \right] dx = \int \frac{dx}{1-x} + \int \frac{dx}{(1-x)^2}$$

$$1-x=u$$

$$-dx=du$$

$$= \int \frac{-du}{u} + \int \frac{-du}{u^2}$$

$$= -\ln|u| - \left(\frac{u^{-1}}{-1} \right) + C$$

$$= -\ln|u| + \frac{1}{u} + C$$

$$= -\ln|1-x| + \frac{1}{1-x} + C$$

$$\text{Ex: } \int \frac{\ln x}{x} dx = ?$$

$$\ln x = u$$

$$\frac{dx}{x} = du$$

$$\begin{aligned} \int \frac{\ln x}{x} dx &= \int u du = \frac{u^2}{2} + C \\ &= \frac{(\ln x)^2}{2} + C \end{aligned}$$

$$\text{Ex: } \int \frac{1}{x \ln x} dx = ?$$

$$\ln x = u$$

$$\frac{dx}{x} = du$$

$$\begin{aligned} \int \frac{dx}{x \ln x} &= \int \frac{du}{u} = \ln |u| + C \\ &= \ln |\ln x| + C \end{aligned}$$

$$\text{Ex: } \int 2x [7 - e^{x^2/4}] dx$$

$$\int (14x - 2x e^{x^2/4}) dx = \int 14x dx - \int 2x e^{x^2/4} dx$$

$$= 14 \frac{x^2}{2} - \int 2 \cdot 2 e^u du = 7x^2 - 4e^u + C$$

$$= 7x^2 - 4e^{x^2/4} + C$$

$$\frac{x^2}{4} = u$$

$$\frac{1}{2} x dx = du$$

$$\text{Ex: } \int \left[\frac{e^{\sqrt{x}}}{\sqrt{x}} + e^2 + x(x^2-16)^2 \right] dx$$

$$\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx + \int e^2 dx + \int x(x^2-16)^2 dx$$

$$x^{1/2} = \sqrt{x} = u$$

$$1/2 x^{-1/2} dx = du$$

$$\frac{1}{2\sqrt{x}} dx = du$$

$$x^2 - 16 = v$$

$$2x dx = dv$$

$$x dx = \frac{dv}{2}$$

$$\int e^u \cdot 2 du + e^2 x + \int v^2 \frac{dv}{2}$$

$$2e^u + e^2 x + \frac{v^3}{6} + C$$

$$2e^{\sqrt{x}} + e^2 x + \frac{1}{6} (x^2 - 16)^3 + C$$

Ex: $\int \frac{1}{\sqrt{x}(\sqrt{x}-2)^3} dx = ?$

$$\int \frac{(\sqrt{x}-2)^{-3}}{\sqrt{x}} dx = \int u^{-3} \cdot 2 du = 2 \cdot \frac{u^{-2}}{-2} + C$$

$$u = \sqrt{x} - 2$$

$$\frac{1}{2\sqrt{x}} dx = du$$

$$\frac{dx}{\sqrt{x}} = 2 du$$

$$= -\frac{1}{u^2} + C$$

$$= -\frac{1}{(\sqrt{x}-2)^2} + C$$

$$e^{\ln a} = a$$

$$a^x = e^{\ln a^x}$$

$$a^x = e^{x \ln a}$$

$$I = \int a^x dx = \int e^{x \ln a} dx$$

$$= \int e^u \frac{du}{\ln a} = \frac{1}{\ln a} e^u + C$$

$$x \ln a = u$$

$$\ln a dx = du$$

$$dx = \frac{du}{\ln a}$$

$$= \frac{1}{\ln a} e^{x \ln a} + C$$

$$\int a^x dx = \frac{1}{\ln a} a^x + C$$

Ex $\int 3^x dx = \frac{1}{\ln 3} 3^x + C$

Ex $\int 5^{1-x} dx = -\int 5^u du = -\frac{1}{\ln 5} 5^u + C$
 $1-x=u$
 $-dx=du$
 $= -\frac{1}{\ln 5} 5^{1-x} + C$

Ex $\int e^{\ln(x^3+5)} dx = \int (x^3+5) dx = \frac{x^4}{4} + 5x + C$

Ex $\int \frac{3^{\ln x}}{x} dx = ?$ $\ln x = u$
 $\frac{dx}{x} = du$

$\int 3^u du = \frac{1}{\ln 3} 3^u + C = \frac{1}{\ln 3} 3^{\ln x} + C$

$$\int \frac{1}{x(\ln x)^2} dx = ?$$

$$u = \ln x$$

$$du = \frac{dx}{x}$$

$$\int \frac{du}{u^2} = \int u^{-2} du = \frac{u^{-1}}{-1} + C$$

$$= -\frac{1}{\ln x} + C$$

Ex: $\int \frac{5+4x^2}{3+2x} dx = ?$

$$\begin{array}{r} -4x^2+5 \quad | \quad 3+2x \\ \underline{+4x^2+6x} \quad -2x+3 \\ 6x+5 \\ \underline{-6x+9} \quad -4 \end{array}$$

$$\int (-2x+3) dx + \int \frac{-4}{2x+3} dx$$

$$-2 \frac{x^2}{2} + 3x - \int \frac{2}{u} du$$

$$-x^2 + 3x - 2 \ln |u| + C$$

$$-x^2 + 3x - 2 \ln |2x+3| + C$$

$$\begin{aligned} 2x+3 &= u \\ 2 dx &= du \\ dx &= \frac{du}{2} \end{aligned}$$

Ex: $\int \left(\frac{1}{x-1} + \frac{1}{(x-1)^2} \right) dx$

$\int \frac{dx}{x-1} + \int \frac{dx}{(x-1)^2} dx$

$x-1=u$
 $dx=du$

$\int \frac{du}{u} + \int u^{-2} du$

$\ln|u| - u^{-1} + C$

$\ln|x-1| - \frac{1}{x-1} + C$

$$\int \frac{2^x}{4^x + 1} dx = ?$$

$$\int \frac{2^x}{(2^x)^2 + 1} dx = \int \frac{du}{\ln 2} \cdot \frac{1}{u^2 + 1}$$

$$\begin{cases} 2^x = u \\ 2^x \cdot \ln 2 dx = du \\ 2^x dx = \frac{du}{\ln 2} \end{cases}$$

$$= \frac{1}{\ln 2} \int \frac{du}{u^2 + 1} = \frac{1}{\ln 2} \arctan u + C$$

$$= \frac{1}{\ln 2} \arctan 2^x + C$$

$$\int \frac{1}{e^x + 1} dx = \int \frac{1 dx}{e^x(1 + e^{-x})} = \int \frac{e^{-x} dx}{1 + e^{-x}}$$

$$u = 1 + e^{-x}$$

$$du = -e^{-x} dx \Rightarrow$$

$$\int \frac{-du}{u} = -\ln|u| + C$$

$$= -\ln|1 + e^{-x}| + C$$

$$\text{Ex: } \int \frac{3\sqrt{1+\tan^{-1}x}}{2+2x^2} dx = \frac{3}{2} \int \sqrt{u} du$$

$$1+\tan^{-1}x = u$$

$$\frac{1}{1+x^2} dx = du$$

$$= \frac{3}{2} \cdot \frac{2}{3} \cdot u^{3/2} + C$$

$$= (1+\tan^{-1}x)^{3/2} + C$$

$$\text{Ex: } \int \frac{e^{\arctan x}}{1+x^2} dx = ?$$

$$u = \arctan x$$

$$x = \tan u$$

$$1 = u'(1 + \tan^2 u)$$

$$\frac{du}{dx} = u' = \frac{1}{1 + \tan^2 u} \Rightarrow du = \frac{dx}{1 + x^2}$$

$$\int \frac{e^{\arctan x}}{1+x^2} dx = \int e^u + C = e^{\arctan x} + C$$